Paper Revision Evidence Audit Pack

Home Energy Tracker — final implementation-and-evaluation revision

Purpose: This companion record explains how the manuscript was revised from the earlier prototype paper. It classifies each important statement by implementation and evidence status, records superseded claims, and identifies what must not be claimed in a presentation or publication.

1. Revision Basis and Title Decision

Source paper reviewed: version-1.1aHome_Energy_Tracker-publication.docx. The requested copy with "(1)" was not found; the matching paper was used. The paper was treated as historical context only. Current source code, Docker configuration, retained dataset/evidence records, and the completed testing guide were used as the source of truth.

TABLE A1. Title candidates evaluated

Candidate	Assessment
A Microservices-Based Home Energy Tracker for Real-Time Monitoring, Intelligent Alerting, and AI-Driven Energy Insights	Rejected: "real-time," "intelligent," and "AI-driven" were too broad and the old paper omitted the evaluated ML subsystem.
A Secure Event-Driven Microservices Home Energy Tracker with ML-Based Anomaly Detection and Deterministic Energy Insights	Not selected: "secure" may overstate local API validation; title is long.
An Event-Driven Microservices Home Energy Tracker with Gateway Authentication and ML-Based Anomaly Detection	Selected: names the demonstrated architectural and ML contributions without asserting physical accuracy, production scale, or universal security.

2. Paper Revision Change Log

TABLE A2. Material claim replacements

Old section / claim	Revision	Reason and evidence
Abstract and introduction: no supervised anomaly detection or validated predictive model.	Describe the evaluated Random Forest and separate offline metrics from runtime integration.	ML training code, model metadata, held-out metrics, and persisted artifact.
Introduction: authentication, authorization, load and resilience testing excluded.	Report bounded gateway security, k6, and service-stop observations.	Security results (8 cases), k6 summary, fault records.
Architecture: gateway, six Java services, Kafka, InfluxDB, MySQL, Mailpit only.	Add Keycloak, device token, ML Inference, anomaly-results, 15-minute deduplication, Prometheus/Grafana, frontend.	docker-compose.yml and service source code.
Alert method: repeated threshold alerts are not deduplicated.	Describe the configured key and 15-minute window; qualify as one controlled functional observation.	AlertService.java and deduplication evidence.
Workflow: no train/test split or empirical classifier evaluation.	Add chronological split, model comparison, held-out results, and integration boundary.	ML code and evidence/01–05.
Functional tests: 19 discovered, 16 executed, 3 skipped.	Separate historical 28 Java/3 skipped, 3 ML API, 4 gateway regression, 8 security, and other experiments by scope/date.	evidence/11 and evidence/15.
Results: 17.32 s analytics and 6.14 s alert observation.	Replace with latest E2E record: 3.79 s analytics availability and 13.64 s threshold-alert/email observation.	Latest successful E2E JSON; explicitly polling-bounded.
Future work: add auth, ML, observability, deduplication, load and fault testing.	Remove as already implemented; retain calibration, provenance, field trials, multi-node testing, and broader failures.	Current implementation/evidence audit.

3. Evidence-to-Paper Matrix

TABLE A3. Claim classification used in the final manuscript

Claim	Paper location	Actual evidence	Status

Claim	Paper location	Actual evidence	Status
17-service local architecture	III-A; IV-A	docker-compose.yml; final validation report	Implemented + functionally observed
JWT roles, ownership, and machine token	III-A; V-C	Gateway source; 8-case security JSON/CSV	Implemented + functionally tested
Time-series usage and deterministic insights	III-B; V-B	Usage/Insight source; latest E2E JSON	Implemented + functionally tested
Random Forest model	III-C; V-A	train.py; joblib; model metadata; metrics	Implemented + experimentally evaluated offline
Kafka ML event path	III-C; V-B	Kafka worker, AlertService, E2E observation	Implemented + runtime observation
15-minute deduplication	III-D; V-D	AlertService; unit and live dedup records	Implemented + functionally tested
Prometheus/Grafana	III-D; V-D	compose/config; dated 8-target snapshot	Implemented + dated runtime snapshot
k6 ingestion behavior	IV-C; V-E	k6 config and JSON summary	Experimentally evaluated locally
Mailpit and ML service-stop behavior	IV-C; V-E	fault records	Controlled manual-restoration observation
Physical-meter / field validity	VI	No matching evidence	Not evaluated

4. Stale-Claim Audit

The following obsolete claims were removed or rewritten. None remain in the revised manuscript.

- No trained classifier or classification metrics.
- No supervised-learning train/test evaluation.
- No Keycloak, JWT, authorization, or ownership enforcement.
- No device/machine authentication for telemetry.
- No alert-deduplication window.
- No observability, Prometheus, or Grafana.
- No load or fault-injection testing.
- Only four-event prototype evidence with 17.32 s / 6.14 s observations as current results.
- Future work to add authentication, ML, deduplication, observability, load testing, and fault injection.

5. Experiment Summary

TABLE A4. Experiment-by-experiment result summary

Experiment	Objective and method	Observed result	Interpretation / limitation
Dataset audit	Inspect CSV schema, quality, timestamp distribution and labels.	5,000 rows, 7 columns, 30-min interval, 95%/5% normal/abnormal; no missing/duplicate/invalid retained rows.	Dataset provenance, licence, label generation, and entity identity are unavailable.
Offline ML	Chronological 70/15/15 split; compare LR and RF; select by validation F1.	RF test F1 0.90244, PR-AUC 0.94853; LR F1 0.10127.	Single retained CSV interval; no field or Wokwi generalization.
Security	Eight gateway-level expected-status API cases.	8/8 expected statuses observed.	Local API validation; neither penetration test nor production-security proof.
Deduplication	Submit duplicate input and inspect alert/log outcome.	One persisted ML anomaly alert plus logged suppression within 15 min.	One functional scenario; no general duplicate-rate claim.
Observability	Inspect Prometheus targets and health endpoints.	8 active targets all up in dated snapshot.	Target reachability only; no long-duration availability or tracing evidence.
k6 load	Local authenticated ingestion ramp to 10 req/s.	450/450 checks; 0 failures; p95 109.77 ms.	One local scenario; no capacity or scalability conclusion.
Fault scenarios	Stop Mailpit/ML service, submit ingestion, manually restart container.	Ingestion 201 during both; recovered=true recorded.	No automated healing, message replay, or end-to-end delivery result.
E2E	Synthetic Wokwi-compatible telemetry with expected analytics and alert assertions.	overallPassed=true; analytics 3.79 s; threshold mail capture 13.64 s.	Synthetic/polling-bounded local functional observations.

6. Reviewer Risk Report

The following are legitimate weaknesses, not defects to conceal. They were retained as limitations or claim boundaries in the revised manuscript.

TABLE A5. Reviewer questions and safe response

Likely question	Evidence-based response / action
Is this a physical smart-meter study?	No. Wokwi and REST records are simulated. The manuscript explicitly avoids meter-accuracy, bill-accuracy, and field-deployment claims.
Are ML metrics valid for Wokwi or homes?	Only for the retained held-out CSV interval. Dataset provenance/labels are undocumented, runtime features are partly imputed, and simulator values may exceed the training range.
Does 450 requests prove scalability?	No. It is a single-host local k6 scenario with a reported p95, not a cloud or distributed capacity study.
Do the service-stop tests prove fault tolerance?	No. They only show accepted ingress during two component stops and manual restoration of the stopped containers.
Does Keycloak prove full security?	No. It demonstrates the eight tested gateway authentication/authorization cases under local conditions.
Does Mailpit prove notification delivery?	It proves local mail-sink capture only, not external SMTP reliability.
Is this a novel algorithm?	No new ML algorithm is claimed. The contribution is evidence-bounded integration and evaluation of existing components.

7. Claims That Must Not Be Made

- “Real-time” without a qualified functional-observation context.
- “Production ready,” “fully secure,” “fault tolerant,” or “highly scalable.”
- Physical-meter accuracy, utility-bill accuracy, appliance-level ground truth, or external email delivery.
- Wokwi or household anomaly-detection accuracy derived from the offline CSV metrics.
- Automatic recovery, Kafka replay, or end-to-end delivery after the recorded fault scenarios.
- ML runtime latency, Kafka lag, distributed tracing, or alert-suppression rates not recorded in the evidence.
- A combined post-fix test total that merges historical and later test suites into one fresh run.

8. Final Quality Gate

Quality gate	Result
Current architecture represented	Pass — Figure 1 and Section III match Compose/source.
ML pipeline and metrics correct	Pass — uses held-out CSV evidence and separate runtime boundary.
Dataset details verified	Pass with limitation — source/licence/label provenance unavailable.
Security/deduplication claims bounded	Pass — describes exact local checks only.
Performance/resilience claims bounded	Pass — local k6 and manual-restoration limits stated.
Old stale claims removed	Pass — audit list reviewed against final content.
Conclusion aligned with results	Pass — local integration/evaluation only.
No fabricated metrics	Pass — every number is from retained evidence or direct arithmetic identified in text.
References	Retained from the existing manuscript; no new unverified bibliographic entry was invented in this revision.